

YOUR OFE 2025 REPORT

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2025 Cropping Season

Field M4F2S2 | NRate 163 lb N/ac | **Field monitoring report — single N rate, no treatment split**

Welcome to your 2025 farm report. This report summarizes field-season measurements for the **M4F2S2** field. Because this field ran under a single nitrogen rate of 163 lb N/ac with no treatment or control strips, we're not comparing two systems here, instead, we're giving you a field-level picture of how the corn crop performed, what the cover crop residue looked like going into the season, and what the soil health assessment tells us about the foundation you're working with. Think of this as your baseline.

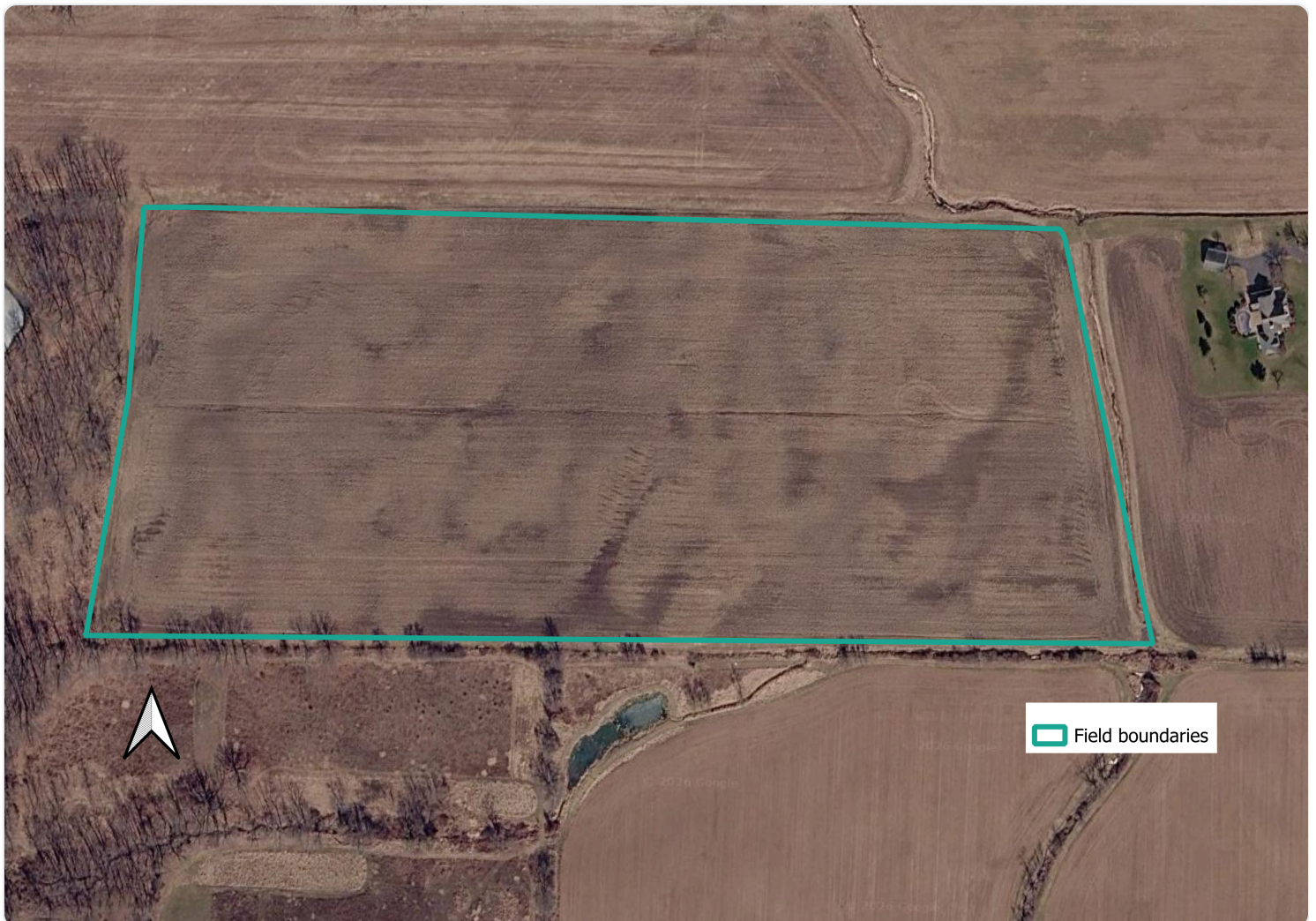


Figure 1. M4F2S2 field.

Take-home message: The corn crop across this field showed high and consistent nitrogen status mid-season, with tissue nitrogen averaging 3.614% across the sampled points at the 163 lb N/ac input rate. The cover crop residue had a C:N ratio of 20.52, which is below 25, so decomposition should be relatively fast and nitrogen should become available early in the season. The soil health assessment returned an overall score of **66/100 (High)**. Chemical indicators are excellent across the board. The main areas to watch are biological and physical structure: soil respiration (24), predicted soil protein (32), and aggregate stability (36) are all in the Low range, and surface and subsurface hardness are at medium functioning. These are the indicators that improve most with reduced tillage and continuous cover cropping over time.

Results Summary

Metric	Value	SD
 Maize Biomass - Mean %N	3.614%	± 0.406
 Maize Biomass - C:N Ratio	12.4	± 1.61
 Cover Crop - Mean %N	2.268%	± 0.115
 Cover Crop - C:N Ratio	20.52	± 1.03
 Soil Health Score	66 / 100	—

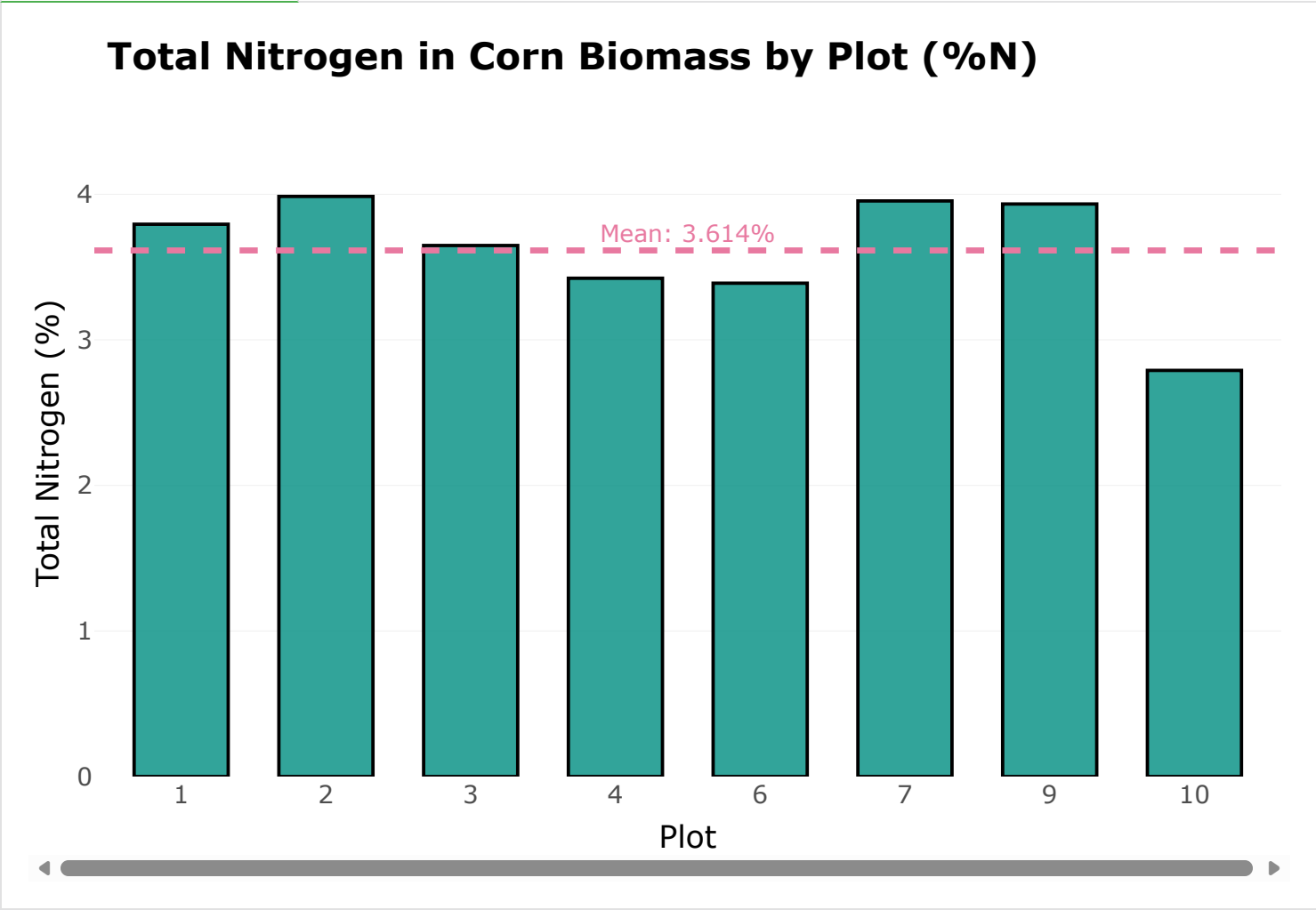
This field did not have a treatment/control split. All statistics are descriptive field-level observations. No statistical comparisons between groups are reported.

C:N in Corn Biomass

Corn biomass was collected from **8 points** across the South field and sent to the laboratory for C:N analysis. This tells us how much nitrogen the plant had absorbed and incorporated into its tissue mid-season, a direct indicator of nitrogen sufficiency at the 163 lb N/ac input rate. Because there is no treatment split, we describe the **field-level distribution and spatial variability** across points.

Total Nitrogen in Biomass (%N)

Bar Chart (%N by plot)



C:N Ratio — Biomass

MEAN C:N — BIOMASS

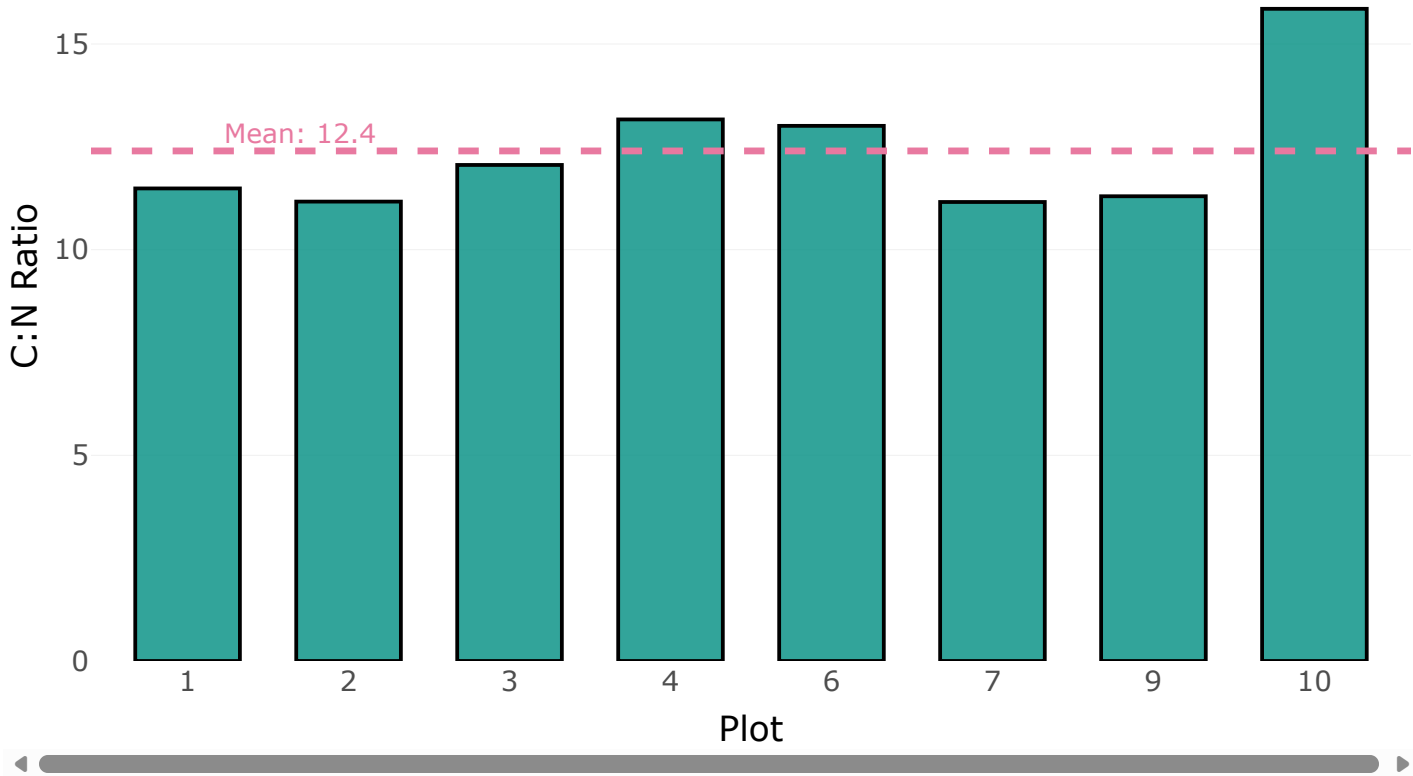
12.4

SD: 1.61 | SE: 0.57

Bar Chart (C:N by plot)

Scatter %C vs %N

C:N Ratio in Corn Biomass by Plot



Cover Crop C:N

5 cover crop samples were collected from field M4F2S2 and analyzed for C:N content. The C:N ratio of cover crop residue tells us how quickly it will break down and release its nitrogen to the following corn crop. A C:N below 25 means decomposition happens relatively fast, the nitrogen becomes available early in the season, right when the young corn plant is establishing its root system and building its nitrogen demand. The bar charts below show the %N and C:N ratio for each sample location.

MEAN %N — COVER CROP

2.268%

SD: 0.115 | n = 5 samples | Range: 2.177 – 2.428%

MEAN C:N — COVER CROP

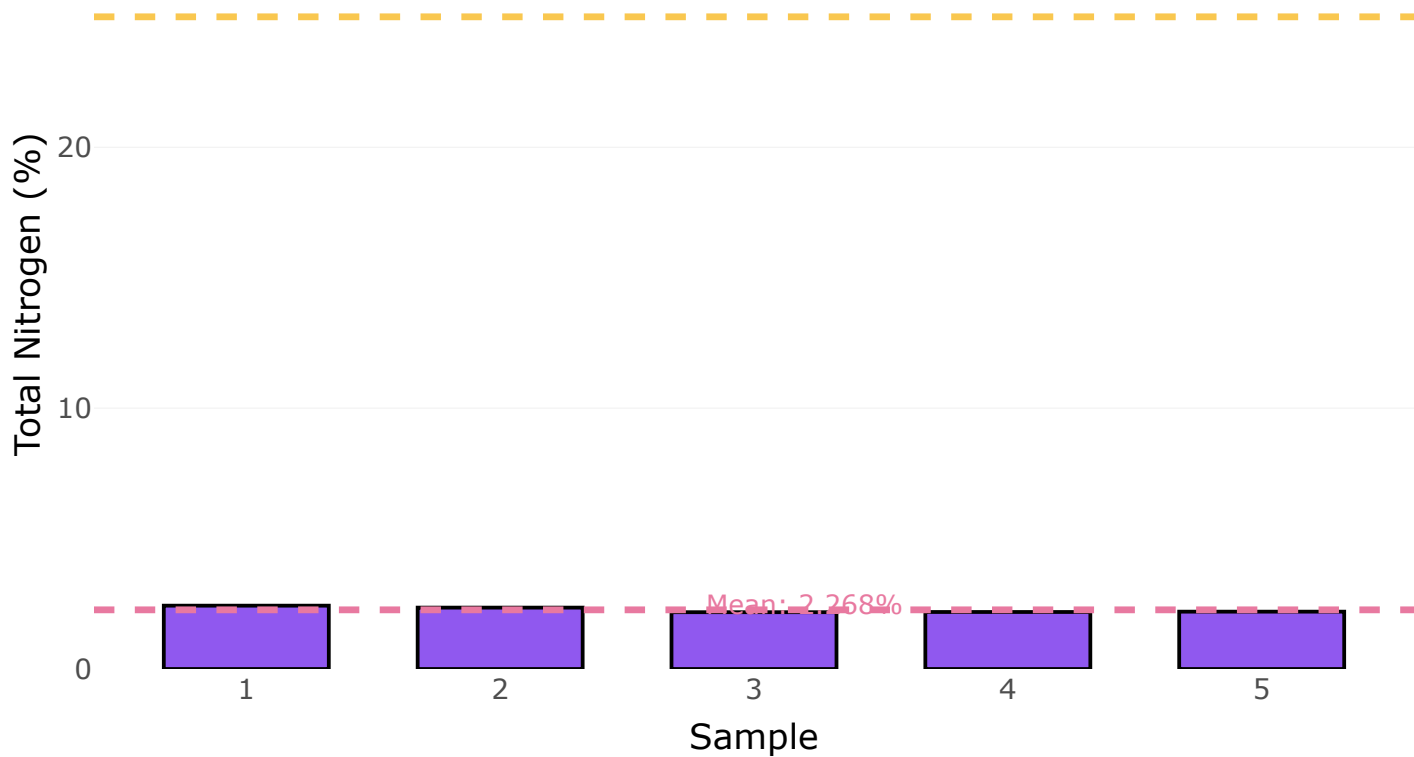
20.52

SD: 1.03 | low C:N — fast decomposition, good early-season N supply

Total Nitrogen in Cover Crop (%N)

C:N Ratio — Cover Crop

Total Nitrogen in Cover Crop (%N)



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **66/100 (High)**. Chemistry is in excellent shape across the board: pH at 6.2 (score 97), extractable phosphorus (86), potassium (100), and all additional nutrients (100). The biological indicators are the main area that needs attention: soil respiration scored 24/100, predicted soil protein 32/100, and aggregate stability 36/100, all in the Low range. These reflect a microbial community that is working below its potential and soil structure that is somewhat fragile under rainfall stress. Surface hardness (56) and subsurface hardness (59) are both in the medium range, indicating moderate compaction worth monitoring. These biological and physical indicators respond best to reduced tillage and continuous cover cropping over time.

OVERALL QUALITY SCORE

66 / 100

Rating: High | Cornell Soil Health Lab

SOIL RESPIRATION

24 / 100

Low — microbial activity below potential

PREDICTED SOIL PROTEIN

32 / 100

Low — N mineralization capacity limited

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: Ron Grushow

Agricultural Service Provider:
Alex Yore
ay327@cornell.edu

Sample ID: QCC90
Field ID: OFE25 N AFTER CC GRUSHOW S
Date Sampled: 07/07/2025
Coordinates: Latitude: 42.881401
Longitude: -77.595345

Measured Soil Textural Class: **silt loam**

Sand: **26%** - Silt: **51%** - Clay: **23%**

Group	Indicator	Value	Rating	Constraints
physical	<u>Predicted</u> Available Water Capacity	0.25	92	
physical	Surface Hardness	151	56	
physical	Subsurface Hardness	274	59	
physical	Aggregate Stability	24.5	36	
biological	Organic Matter Soil Organic Carbon: 1.76 / Total Carbon: 1.81 / Total Nitrogen: 0.18	3.2	59	
biological	<u>Predicted</u> Soil Protein	4.99	32	
biological	Soil Respiration	0.4	24	
biological	Active Carbon	507	52	
chemical	Soil pH	6.2	97	
chemical	Extractable Phosphorus	3	86	
chemical	Extractable Potassium	109.4	100	
chemical	Additional Nutrients Ca: 1428.1 / Mg: 222.9 / S: 6.1 Al: 15.2 / B: 0.12 / Cu: 0.1 Fe: 2.1 / Mn: 3.3 / Zn: 1.2		100	

Overall Quality Score: **66** / High

Figure 2. Cornell Assessment of Soil Health report - M4F2S2.

[Open Full Soil Health Report \(PDF\)](#)

Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index from June through November 2025. A large single-day rain event hit in mid-June, around 65 mm, which pushed your cumulative total well above the all-site average heading into July. From there, rainfall stayed consistent and moderate through the summer, tracking closely with the network average. The drought index stayed at None through most of September. It moved to Low in late September and to Mild in October, and your field finished the season slightly below the all-site average (about 425 mm vs 470 mm). One thing worth watching: given the low aggregate stability score (36/100), that heavy June rain event may have contributed to some surface crusting or runoff, something to monitor in future seasons, particularly given the silt loam texture of this field.

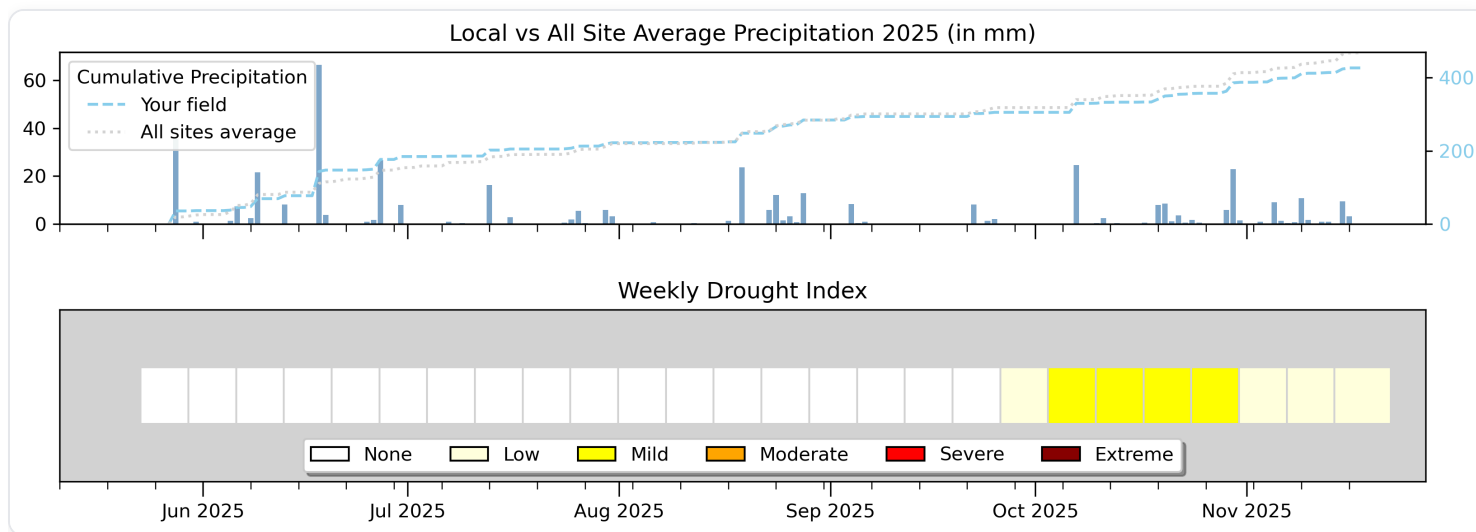


Figure 3. Local vs all-site average precipitation (top) and weekly drought index (bottom), June – November 2025.

Conclusions

MAIZE BIOMASS NITROGEN

Corn plants across the field averaged **3.614% N** in mid-season biomass, with a C:N ratio of **12.4**. Tissue nitrogen was notably higher than in the M5F2S2 field, which received the same rate, across the 8 sampled locations at the 163 lb N/ac input rate.

COVER CROP N CONTRIBUTION

The cover crop came in at a mean C:N of **20.52** and tissue N of **2.268%**, a low C:N ratio (below 25), which points to relatively fast decomposition and early-season nitrogen release. The five samples were very similar to each other.

BIOLOGICAL ACTIVITY AND AGGREGATE STABILITY — PRIORITY CONSTRAINTS

Soil respiration scored **24/100** and predicted soil protein **32/100**, both in the Low range. Aggregate stability (36/100) is also low, meaning soil structure is somewhat vulnerable to rainfall-driven disruption. The microbial community is working below its potential. Maintaining plant cover, adding fresh organic matter, and reducing tillage intensity are the primary levers to move these numbers up over time.